

- Improved iron ore availability in Karnataka after mines restart operation following implementation of Resettlement and Rehabilitation (R&R) schemes;
- Continued downfall in iron ore output and exports from Goa and Odisha at the back of allegations of illegal mining.

4.7.3 Acquiring Mineral Sources Abroad

In view of the worsening commodity-crisis, the government has been advised to adopt the Chinese Model of buying foreign minerals and mining rights. The principal focus in the Chinese Model is on acquisition of ownership of reserves in coal, oil and gas.

Acquiring mineral reserve in other resource-rich nations is considered key to India's energy security, owing to domestic constraints in exploiting reserves and the global rise in commodity prices. Also, holding such assets acts as a strategic tool in energy diplomacy for developing and maintaining foreign relations.

4.8 ALLOCATION OF NATURAL RESOURCES

With growth gaining pace not only in India but in many emerging economies, the pressure on natural resources is on the rise. Innovations and technological progress have opened up undreamt vistas of use of natural resources. This in turn implies unprecedented opportunities to make private profit from publicly-owned goods.

Crony capitalism, which involves both corrupt politicians and unscrupulous businessmen in a close embrace to manipulate and use the discretionary powers vested in government for private profit. The allocation of publicly controlled natural resources and land is the biggest playground for this collusion.

In 2012 judgement on 2G case, the Supreme Court made a serious attempt to control such collusion; it has made efforts to ensure that the resource rents should go to the public treasury.

There are different methods to distribute scarce resources like (i) first-come-first-served method, (ii) auction, and (iii) open access, etc.

- i) **First-come-first-served:** In the first-come-first-served method all the intending buyers are put in a queue. The available resources are allotted to an entity as per the sequence in the queue. The process is complete when the last available unit has been allocated. On the face of it this method appears both transparent and non-discriminatory, and hence worth adopting. But a major shortcoming of this method is that in our system, somebody or the other will always get or, more likely be given, advance information so that they can position themselves up ahead in the queue.
- ii) **Auction or competitive bidding:** The Supreme Court (SC), like the Ashok Chawla committee earlier, has come out strongly in favour of auction method of allocating the right to publicly controlled resources. The SC in its order said: "When it comes to alienation of scarce natural resources like spectrum etc. the state must always adopt a method of auction by giving wide publicity so that all eligible persons may participate in the process. Auctions or

competitive bidding would be the more appropriate method as it is likely to lead to allocation of resources to those who can use them most efficiently.

But this method also suffers from various limitations. (i) Designing an auction requires prior consideration about how many licenses should share the resource to ensure an adequate level of competition— and that is a judgement call that can be distorted by ulterior motives. (ii) Any workable auction will also have to have some qualifying criteria on who is eligible to compete and that is even more liable to manipulation. (iii) The experience shows that disclosure requirements are often lax which allows interested parties to corner resources through front companies. (iv) This method can be used when there are technical or logistical reasons for apportioning and giving licensees *de facto* ownership rights. But that is not always necessary. The Supreme Court has also held this view.

- iii) **Open Access.** An alternative is to provide open access to the resources to all eligible users against a usage charge, which could be calibrated to ensure that the demands on the resources match availability. Open access with pay-per-use is better for maintaining competition and easing entry by new players.

Open access will not work when the resource deplete with overuse. It would lead to what has been described as the tragedy of commons, where it is in the commercial interest of each user to extract as much as possible without regard to optimum rates of exploitation. When a resource can be over-exploited, the right of access to the resource has to be organised so as to create checks for staying within sustainable yields or by setting up barriers to excessive exploitation. One approach, typically followed in mineral and oil concessions, is to give the licensee long-term rights over a well-defined resource block. In principle, the licensee should be interested in avoiding unsustainable rates of exploitation that reduce the total return over time.

Another approach, typically followed in forestry, is to give short-term rights of access with controls that specify the amounts that can be extracted. This approach depends on honest and effective policing, and the evidence from the past history of forestry in India does not give any grounds for confidence that it would work ultimately, the goal must be to eliminate the discretionary powers of politicians and bureaucrats to choose concessionaries, instead they should be chosen through a system of public auctions, both for outright transfer of rights or for public-private appropriation of resource rents, ensure free and fair competition and create incentives for each user to stay within the limitations of optimal exploitation. It is necessary that the competition for natural resource markets, must be transformed to competition in those markets. A truly transparent and honest system will require some independent scrutiny of the design and the rulers of process that all resource concessionaries should follow.

Check Your Progress 2

- 1) What are the major issues related to biodiversity and forestry that India must address in keeping with its current needs and future objectives?

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- 2) Why should India have a policy to acquire mineral resources located abroad?

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- 3) Examine the different methods used in natural resource allocation. Based on your understanding and India's recent experience in undertaking those allocations, suggest the preferred approach in that regard.

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4.9 ENVIRONMENT AND ECONOMIC DEVELOPMENT

Environment has been defined as the aggregate of all the external conditions and influences affecting the life and development of an organism. It comprises the whole range of external influences acting on an organism, of both the physical and the biological forces of nature surrounding the human individual. All living beings are part of balanced and interacting eco-system: they draw sustenance from the solid, liquid, and aerial resources on the earth. They undergo passage through the phases of reproduction, sustenance, and extinction. The ultimate source of energy for the entire ecosystem is the solar energy. The ecosystem consists of subsystems such as sea, forests, water reservoirs, plants, trees, insects, animals in a forest– all of them are inter-connected in a network of devourer-devoured relationship. None of the elements in the network is useless or purposeless, each one of them helps to regulate the balance of the subsystem. Bio-gas-chemical cycles are essential for the process of reproduction in the physical environment. The process of reproduction is in balance as long as the natural cycles, their interrelations and the hierarchy of the food-chain are not disturbed.

There are many different models of the relationship between the environment and the economy. The simple model depicted in Figure.4.3 illustrates the four functions of the environment in supporting economic activity and the effects of this activity on the environment. These four functions are life-support, supply of natural resources, absorption of waste products and supply of amenity services. The economy is represented in Figure 4.3 by households consuming goods and services, and firms producing with natural resources provided by the environment, with labour and man-made capital provided by households.